



From Scratch to Python

Lesson 2



Lesson Objectives

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- Remember how to program outputs in python,
- Remember how to program inputs in python,
- Remember the need to store inputs in python,
- Remember how to store inputs into variables in python
- Understand the difference between strings and variables
- Understand how to concatenate strings and variables.

Lesson Outcomes

- Carrying out a number of programming tasks.

Literacy – Key Words

Input	Values which get sent from the user into the computer
Variable	The place where inputs get stored by the program
Output	The values which get sent from the computer to the user



Can you remember the following...

The Python Code needed to display some words on the screen?



...in Python we use **print()**

In python:

Output
↓

<pre>1 print("hello") 2 3 4</pre>	<pre>hello</pre>
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The print() statement will output the contents of its brackets.

If we wish it to print some text, the text must be surrounded by quotes!



Can you remember the following...

The Python Code needed to ask the user for an input?



Programming Inputs & Storing Inputs in PYTHON

In python:

```
1 answer = input("What is your name?")
2 print(answer)
3
```



coder.computerscienceuk.com says

What is your name?

OK Cancel



In Python, the 'input' statement would allow our program to ask the user to enter an input.

The 'answer' variable would store the input.

And we could use the answer variable with a 'print' statement to output the inputted value.

Output



```
1 answer = input("What is your name?")
2 print(answer)
3
```

Mr Wickins



Can you tell me the following?

1. The python code needed to display a word on the screen? 

```
1 print("hello")  
2
```

hello

2. The python code needed to ask the user for an input? 

```
input("What is your name?")
```

3. The way python can store user inputs? 

```
answer = input("What is your name?")
```



What is the Major Difference between a print and an input statement?

```
1 print("hello")
```

```
hello
```

PRINT Just displays a message!

```
input("What is your name?")
```

INPUT Displays a message and waits for a user input!

```
1 answer = input("What is your name?")
```

```
Mr Wickins
```

```
2 print(answer)
```

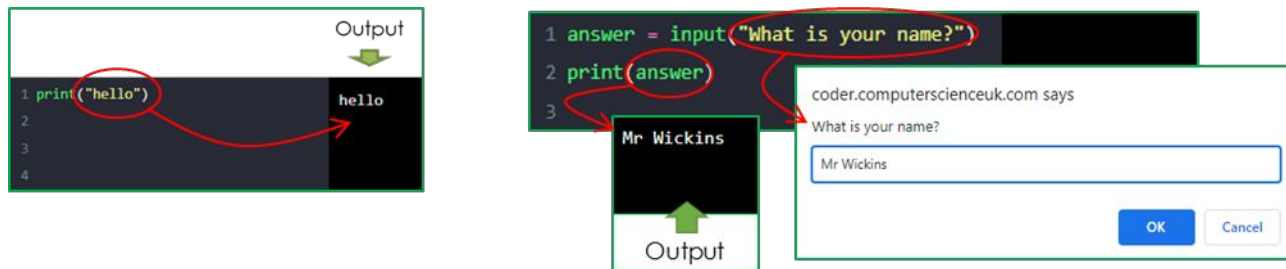
```
3
```

So, it needs to equal a variable if we want to store what the user types in...



Strings, Variables, Quotes and Concatenation

So far, we've seen how `print()` and `input()` statements can display text on the screen.



However, it's important to recognise...

1. The difference between outputting text and outputting the contents of variables
2. How text and variable contents can be joined together in `print()` and `input()` statements.

So let's look at `print()` and `input()` statements in a little more detail.



Strings, Variables and Quotes

In programming, **text** is known as '**strings**'. A '**string**' is a data type that consists of various keyboard characters.

Variables are memory locations (storage areas) which can contain strings (text).

When we wish to **output strings** in a `print()` or `input()` statement, we must use quotes. For example, the following code...

```
print("Bob")
```

...would output the *string* **Bob** onto the screen.

If we don't use quotes, python will assume that we wish to **output a variable** of that name. For example, the following code...

```
print(Bob)
```

...would cause the program to look for a *variable* called **Bob** and output its contents. And if the variable doesn't exist, the program would output an error.



Concatenation

In python, to join a string and a variable (in a print() or input() statement), to form a long sentence, we can use the '+' operator.

```
1 name = "Jimmy"
2 print("Hello " + name + ", it's good to meet you!")
```

Hello Jimmy, it's good to meet you!

Above, you can see that the program has a variable called 'name', which contains the string 'Jimmy'.

In the print() statement underneath, the '+' operator has been used to join (concatenate) the string "Hello ", with the contents of the variable, and with another string ", it's good to meet you!", to form the sentence "Hello Jimmy, it's good to meet you!".

When using the '+' operator to form sentences, remember to add spaces in your strings in the appropriate positions, otherwise two words may be joined together in your sentence.



Task 1 – Lets get programming!

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Program 1

Write a program that will allow the user to input their name and age and then output the name and age that have been entered on the screen.

Program 2

Write a program that will allow the user to input their name and then output the name joined with a greeting in the format: "Hello Joe".

15 mins



What is happening here?

Write your thoughts on a whiteboard

This input statement asks the user a question and waits for their input

The input is then stored in a variable called **ict**

```
• ict = input("What is your current level in ICT?")
• maths = input("What is your current level in Maths?")
• geog = input("What is your current level in Geography?")

• print("Below is a summary of your current levels: ")
• print("")
• print("ICT Level: ", ict)
• print("Maths Level: ", maths)
• print("Geography Level: ", geog)
• print("")
• print("...good luck improving those!")
```

This is repeated
for the other
subjects

Finally each print
statement outputs
different lines of text

Here the text in speech marks is output. And the contents in variables too. Speech marks aren't needed here otherwise it would output the text ict instead of the variables contents.



Task 2 – Lets get programming!

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Program:

Write a program, similar to the one just shown, which does the following:

- Asks for your current level in all of your subjects
- Stores the inputs in different, suitably named variables
- Displays your current levels in an organised, well presented report on the screen

15 mins



Lesson Activity

- Read the information in the lesson's workbook
- Complete the PRIMM tasks provided.

